Improving Policy Learning via Language-Guided State Representation in World Models

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***Abstract*—World models have emerged as powerful technology for facilitating policy learning in robotics by providing predictive representations of environmental dynamics. A crucial component of such models is the internal state representation, which serves as a bridge between observation, decision-making, and future state prediction. However, in multi-task settings, the presence of task-irrelevant information within the representation, especially under limited information capacity, can significantly impair the effectiveness of policy learning. To address this challenge, we propose Language-Guided State Representation (LGSR), a novel method that leverages language-based task descriptions as priors to guide the encoding of state representations within the world model. By incorporating language instructions, LGSR promotes the extraction of task-relevant features while suppressing irrelevant information, thereby enhancing the alignment between representation learning and task objectives. We evaluate LGSR on the CALVIN benchmark, a suite of language-conditioned robotic manipulation tasks. Experimental results demonstrate that incorporating language priors into the state representation enhances the extraction of task-relevant information, thereby significantly improving the efficiency and success rate of downstream policy learning.**

***Index Terms*—Representation Learning, Imitation Learning, Learning from Demonstration**

I. INTRODUCTION

HUMANS naturally construct an abstract model of the real world, encoding compact temporal and spatial information in their minds to enable fast and efficient decision-making [1]. Inspired by this cognitive ability, world models aim to learn compact and predictive representations of environmental dynamics, allowing agents to simulate future states internally for efficient robotics policy learning [2]. Typically, a world model consists of an encoder that compresses high-dimensional perceptual observations, a dynamics model that captures environment transitions, and several prediction heads that reconstruct the properties of the environment, such as object semantics [3], depth maps [4], or optical flow [5]. Among these components, the latent state representation serves as a crucial bridge between perception and decision-making, providing a compact and structured space where policies can be optimized with fewer environment interactions. Consequently, the quality of state representation learning constitutes a key bottleneck in enhancing the effectiveness of world models for policy learning.

Different from conventional representation learning methods in computer vision [6], [7] and natural language processing [8], [9], the world model learns representations by predicting future states conditioned on various actions. By modeling future changes under different actions, the state representation is implicitly encouraged to capture critical features of the environment [10]. For example, in a block-pushing task, predicting the future position of the block requires the model to capture latent physical properties such as friction and contact dynamics, enabling a compact representation that facilitates more efficient policy learning of the block-pushing task [11]. Based on this principle, recent advances in world models emphasize learning a compact state representation followed by policy learning. GenAD [12] introduces a lightweight planner on diffusion features from future video prediction to generate waypoints within minutes. ManiGaussian [13] predicts actions from spatially enhanced Gaussian point representations that capture semantic propagation. Seer [14] predicts future states from task instructions and generates executable actions with a language-conditioned world model. Together, these studies show that effective state representations significantly improve the performance and sample efficiency of policy learning.

However, in multi-task policy learning [15], where a single policy must solve diverse tasks with varying goals, standard world models from previous studies focus solely on encoding raw observations into latent state representation, while ignoring the role of task language descriptions in guiding state representation learning. This limitation becomes especially severe in multi-task settings, where a single encoder often suffers from semantic entanglement and limited capacity to isolate task-relevant features. In this context, task language descriptions typically provide goal-related priors, such as the target object and its location, which can help the model focus on critical entities in the environment. Without leveraging these priors, the model may encode task-irrelevant details, reducing policy learning efficiency under limited representation capacity. For instance, in a task of picking red cubes from a cluttered table, encoding irrelevant objects (e.g., a green ball) may obscure the essential features of the red cubes.

To address the absence of task-informed guidance in state representation learning for world models, we propose

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Language-Guided State Representation (LGSR), which leverages language-based task descriptions to promote the task-relevant state representation learning within world models. Specifically, we encode language descriptions and visual observations with pre-trained foundation models and integrate them through a multimodal fusion module. To further encourage the state representation to focus on task-relevant features, we introduce an auxiliary training objective based on normalized time-to-go prediction, which explicitly aligns representation learning with task completion. With these designs, LGSR enhances the extraction of task-relevant features while suppressing irrelevant information, thereby achieving better alignment between representation learning and task objectives, and significantly improving the multi-task policy learning performance.

Briefly, the main contributions of this work are as follows:

- We propose Language-Guided State Representation (LGSR) for world models in multi-task policy learning, which leverages the prior information in task descriptions to guide the state representation learning by extracting the task-relevant and suppressing the task-irrelevant information, thereby improving the multi-task policy learning performance.
- To better incorporate the prior information in the task description into state representation, we design a normalized time-to-go signal as the proxy reward to encourage the state representation to encode information critical for achieving task objectives.
- Superior performance is achieved in various language-conditioned robotics manipulation tasks from the CALVIN benchmark, demonstrating the effectiveness of our method in improving multi-task policy learning through language-guided state representation learning.

II. RELATED WORK

This section begins with an overview of recent advances in world models for robot learning. Next, we review several representative approaches to language-conditioned policy learning. Finally, we investigate methods for task-relevant representation learning in robotics.

A. World Models in Robot Policy Learning

World models have shown great potential for improving robot policy learning and have attracted increasing attention in robotics. Their applications mainly span RL sample-efficiency improvement, online planning with learned predictive models, and imitation learning with enhanced state representations. For RL, DreamerV1–V3 [2] learn latent dynamics models to simulate environment interactions and improve sample efficiency. Later studies replace reconstruction prediction with reward prediction [16] or introduce contrastive learning [17], yielding more compact states and better locomotion and manipulation performance. For online planning, Pathdreamer [3] predicts future observations from sampled trajectories and selects the optimal one using an instruction-trajectory compatibility reward, with similar ideas applied to vision-and-language navigation [18] and autonomous driving [19]. In imitation learning, prior work improves performance and sample efficiency via enhanced state representations, such as diffusion features [12], Gaussian Splatting embeddings [13], predictive coding [20], and future state prediction [21], [14]. However, most world models still learn task-agnostic state representations without explicit task-intent conditioning, limiting their effectiveness in multi-task settings. We therefore propose language-guided state representation learning for multi-task imitation learning.

B. Language-Conditioned Policy Learning

Language-conditioned policy learning aims to enable agents to execute diverse natural-language tasks and has been extensively studied [22], [23]. MCIL [24] learns a shared latent goal space between language and image goals, leveraging both language-annotated and language-free data, and serves as the CALVIN benchmark baseline [15]. Related benchmarks such as LIBERO [25] further emphasize lifelong knowledge transfer across diverse manipulation tasks. Subsequent works extend vision-language models to policy learning, such as Roboflamingo [26], which improves object grounding, language understanding, and vision-language alignment. PerAct [27] adopts a Perceiver Transformer with query-based fusion to process 3D voxel observations and predict manipulation actions, while similar mechanisms also appear in vision-language models such as InstructBLIP [28]. Hierarchical methods, including SuSIE [29], CLOVER [30], and GHIL-Glue [31], predict language-conditioned future observations and infer actions using inverse dynamics models. GR-1 [32] improves policy learning through generative video pretraining, while GR-MG [33] leverages multimodal goals and partially annotated data for more efficient learning. Seer [14] further builds a language-conditioned world model to predict future states and generate executable actions. Despite their effectiveness, these methods mainly use language at the policy or prediction level, while its potential as a prior for state representation learning remains underexplored. We address this gap by learning language-guided state representations within world models.

C. Task-Relevant Representation Learning

High-dimensional observations usually contain redundant information that does not contribute to decision-making. This motivates research on learning concise, task-relevant representations in decision-making tasks. Nakanoya et al. [34] compress high-bitrate sensory data into task-relevant representations for bandwidth-constrained cloud applications. Yamada et al. [35] learn task-induced representations using task rewards and demonstrations to suppress distractors, while Yang et al. [36] extract task-relevant representations by modeling reward sequence distributions. MVRL [37] applies bisimulation metric learning to multi-view observations, and Kim et al. [38] learn task-relevant features by predicting observations with masked distractors. Although these methods effectively learn task-relevant representations, they largely overlook language descriptions as guidance for state representation learning. The closest work to ours is CARE [39], which uses task metadata to extract task-relevant representations. However, CARE focuses on composable cross-task representations to reduce negative transfer in reinforcement learning, whereas we leverage prior knowledge from task descriptions to learn task-specific state representations, distinguishing our approach from prior efforts.

III. PROBLEM FORMULATION

In this work, we formulate multi-task policy learning as a language-conditioned robot control problem, which can be modeled as a goal-conditioned partially observable Markov decision process (GC-POMDP) [40], defined as $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{T}, \rho_0, \mathcal{L}, \phi)$, where $\mathcal{S}$ and $\mathcal{O}$ denote the set of states and observations, respectively; $\mathcal{A}$ is the action space; $\mathcal{T}$ is the environment transition dynamics; ρ_0 is the initial state distribution; $\mathcal{L}$ represents the set of free-form language instructions; $\phi(s)$ is a binary function indicating task success. Specifically, at the beginning of each episode, the robot is given a language instruction $l \in \mathcal{L}$, which remains fixed throughout the episode. At every time step t, the observation o_t includes proprioceptive readings and multi-view images from a third-perspective camera and a gripper camera. The control policy is modeled as a language-conditioned policy $\pi(a|o, l)$, where the action space is typically specified as the relative Cartesian displacement of the gripper and its open/close status. The learning objective of the language-conditioned robot control problem in our imitation learning setting is to maximize the expected task success rate across N episodes $\mathbb{E}[\phi(s_T)] = \frac{1}{N}\sum_{i=1}^{N} \phi(s_T^i)$. Since ϕ is sparse and non-differentiable, we follow standard imitation learning and optimize a behavior-cloning surrogate over expert demonstrations as the primary objective in Section IV-B3.

To support efficient multi-task policy learning, we adopt a joint world model and policy learning framework. In this framework, the world model $p(s_{t+1} \mid s_t, l)$ captures the language-conditioned dynamics in latent space, where s_t denotes the compact state representation. The policy $\pi(a_t \mid s_t)$ is optimized based on this learned representation. In conventional approaches, s_t is typically inferred solely from o_t, resulting in task-agnostic or single-task-specific representations that may encode redundant or irrelevant information. This becomes particularly problematic under limited representation capacity, where critical task-relevant features can be overlooked. To address this, we propose to incorporate the task instruction l into the state representation learning process, thereby guiding world models to learn task-relevant state representations that better align with task objectives.

IV. METHODS

In this section, we first introduce the multimodal perceiver encoder (Section IV-A), which produces the Language-Guided State Representation (LGSR) that serves as the core of our method. We then describe the joint world model and policy learning framework (Section IV-B), which leverages LGSR to enable efficient multi-task policy learning.

A. Language-Guided State Representation

As illustrated in Fig. 1, the multimodal perceiver encoder (MPE) receives the task language description l and the visual observation o_t at each time step to produce the language-guided state representation s_{lgsr}:

$$s_{lgsr} = \mathrm{MPE}(o_t, l). \tag{1}$$

1) Feature Extraction with Foundation Models: To effectively extract semantic knowledge from both language and visual inputs, we utilize pre-trained foundation models as strong perceptual encoders to encode the raw data into informative embeddings. For the language inputs, we tokenize the instruction and use the CLIP text encoder [41] to obtain text embeddings. For the visual input, we patchify the image and apply a pre-trained masked autoencoder (MAE) [6] encoder to extract visual embeddings. The MAE encoder effectively extracts high-level semantic information from the image, which is particularly beneficial for capturing object-level attributes and contextual relations in the scene.

$$e_{\text{text}} = \mathrm{CLIP}(l), \quad e_{\text{vis}} = \mathrm{MAE}(o_t). \tag{2}$$

2) Multimodal Fusion via Perceiver Resampler: To enable the language instruction to guide task-relevant visual feature extraction, we fuse these two modalities using a perceiver resampler (PRS) [42]. The PRS maintains a set of learnable latent tokens, which interact with both language and visual embeddings through multiple layers of cross attention. During this process, the latent tokens absorb task-relevant priors from the language, which in turn modulate the attention to visual features, resulting in language-guided state representation:

$$s_{lgsr} = \mathrm{PRS}([e_{\text{text}}, e_{\text{vis}}]). \tag{3}$$

3) Reward-enhanced Representation Learning: Building upon the structural injection mechanism provided by the perceiver resampler for language guidance, we further introduce an auxiliary time-to-go prediction objective that complements the language-fusion mechanism with additional temporal supervision. Specifically, we define a normalized time-to-go target r that decreases monotonically from 1 at the first frame to 0 at the success frame along a successful demonstration:

$$r \triangleq (T - t)/T, \tag{4}$$

where T denotes the last-frame index of each trajectory. The target is obtained from successful demonstrations without additional annotations. We train a reward head (RH) to predict this proxy reward from LGSR s_{lgsr}:

$$\hat{r} = \mathrm{RH}(s_{lgsr}), \tag{5}$$

using the mean squared error (MSE) loss:

$$\mathcal{L}_r = \mathrm{MSE}(\hat{r}, r). \tag{6}$$

This auxiliary loss encourages the state representation to capture task-related features, complementing the language-fusion mechanism.

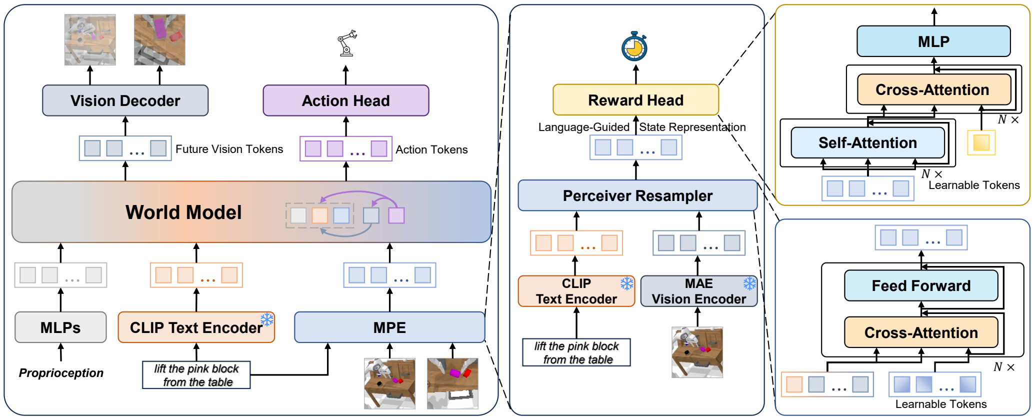


Fig. 1. Illustration of the proposed framework. Our method integrates a world model with a Language-Guided State Representation (LGSR) learning module. Visual observations and language instructions are encoded by pre-trained MAE and CLIP, respectively, and fused via a multimodal perceiver encoder (MPE) to form the LGSR. The LGSR is then used by the world model to predict future visual observations and actions jointly. A reward head further complements LGSR learning with task progress supervision.

B. Joint World Model and Policy Learning

As shown in Fig. 1, we adopt a joint world model and policy learning framework to facilitate multi-task policy learning. Similar to previous work [13], [14], we jointly optimize future state and action predictions using a transformer-based neural network. This approach enables efficient learning of both predictive and task-relevant representations, improving performance across diverse tasks.

1) Feature Extraction and Encoding: At each time step t, the inputs of the world model consist of the language instruction l and the observation o_t, which includes proprioceptive reading o_t^{prop}, and multi-view images o_t^{vis} from a third-perspective camera and a gripper camera. For the language instruction, we first tokenize the instruction and convert it into text embeddings using the CLIP text encoder [41]. For the visual inputs, we extract task-relevant features from both the third-perspective and gripper images via the MPE encoder, as described in Section IV-A. The proprioceptive readings are encoded into tokens with the same embedding dimension through a multi-layer perceptron (MLP) network.

$$e_{\text{text}} = \text{CLIP}(l), e_{\text{vis}} = \text{MPE}(o_t^{\text{vis}}), e_{\text{prop}} = \text{MLP}(o_t^{\text{prop}}). \tag{7}$$

Additionally, to enhance the robustness of the policy and capture long-range dependencies, we include the observations from previous H time steps as history $h_t = \{o_{t-H}, \dots, o_{t-2}, o_{t-1}\}$ in the world model's input. For early time steps, we replicate the earliest available observation to fill missing history slots, keeping the input length fixed.

2) World Model and Policy Learning Architecture: For joint world-model and policy learning, we employ a transformer-based masked prediction architecture inspired by BERT-style masked modeling [8]. In this framework, target elements are replaced with learnable mask tokens, and the model learns to recover them from the remaining inputs. Specifically, the future observation o_{t+N} and the chunked action $a_{t:t+M}$ [43] serve only as decoding targets, while learnable mask tokens m_o and m_a are inserted into the input sequence as their prediction slots. Finally, we organize the inputs and outputs as follows:

$$[\, h_t,\ l,\ o_t,\ m_o,\ m_a \,]. \tag{8}$$

As shown in (8), to model future state prediction in the world model (WM), we replace the future observation o_{t+N} with learnable future-observation tokens m_o. Under the attention mask, these tokens aggregate information from the available contextual inputs, including the encoded history e_{his}, language embedding e_{text}, visual embedding e_{vis}, and proprioceptive embedding e_{prop}. The hidden states at these positions form the latent future-observation representation:

$$e_f = \text{WM}(e_{\text{his}}, e_{\text{text}}, e_{\text{vis}}, e_{\text{prop}}, m_o), \tag{9}$$

where e_{his} refers to the encoded history observation tokens.

For policy learning (PL), the chunked actions $a_{t:t+M}$ are replaced by learnable action tokens m_a. These tokens attend to the contextual inputs and to the latent future-observation representation e_f, yielding the action representation:

$$e_a = \text{PL}(e_{\text{his}}, e_{\text{text}}, e_{\text{vis}}, e_{\text{prop}}, e_f, m_a). \tag{10}$$

Here, WM and PL share the same transformer, where m_o and m_a serve as prediction slots for future-observation and action prediction, respectively. This design allows the representations e_f and e_a to be produced in a single forward pass, with e_a conditioned on e_f through layer-wise attention.

3) Future Prediction and Action Learning: To optimize the world model and action prediction, we introduce a vision decoder (VD) to predict future observations and an action head (AH) to decode the chunked actions:

$$\hat{o}_{t+N} = \text{VD}(e_f), \quad \hat{a}_{t:t+M} = \text{AH}(e_a). \tag{11}$$

Specifically, since the primary goal of world model learning is to improve policy learning through predictive representations, we focus on capturing the environment dynamics rather than reconstructing the original observation details. To achieve this, we normalize the original visual observation with patch-wise

TABLE I
TASK IDENTIFICATION ACCURACY OF LEARNED REPRESENTATIONS.

Variant	CALVIN ABC-D	LIBERO-Long
Seer	$48.54 \pm 1.19\%$	$99.62 \pm 0.15\%$
LGSR	$97.94 \pm 0.32\%$	$100.00 \pm 0.00\%$
LGSR $+\mathcal{L}_r$	$98.38 \pm 0.41\%$	$100.00 \pm 0.00\%$

zero-mean normalization (ZMN) and use the mean squared error (MSE) loss:

$$\mathcal{L}_f = \mathrm{MSE}\left(\hat{o}_{t+N}, \mathrm{ZMN}(o_{t+N})\right). \tag{12}$$

For action prediction, we separate the action space into continuous and discrete parts for relative Cartesian displacement and open/close status, respectively, and use MSE and cross-entropy (CE) loss for them as follows:

$$\mathcal{L}_a = \mathrm{MSE}(\hat{a}^{\mathrm{cont}}_{t:t+M}, a^{\mathrm{cont}}_{t:t+M}) + w_{\mathrm{disc}}\mathrm{CE}(\hat{a}^{\mathrm{disc}}_{t:t+M}, a^{\mathrm{disc}}_{t:t+M}), \tag{13}$$

where w_{disc} is the loss weight for discrete part. The final loss function is the weighted sum of these components:

$$\mathcal{L} = w_a\mathcal{L}_a + w_f\mathcal{L}_f + w_r\mathcal{L}_r, \tag{14}$$

where the w_a, w_f, w_r are the loss weights for action, future observation, and reward prediction losses.

V. EXPERIMENTS

We evaluate the proposed Language-Guided State Representation (LGSR) approach on the CALVIN benchmark [15] to address the following research questions:

- **Effectiveness:** Can LGSR improve multi-task policy learning performance compared to baseline methods?
- **Data Efficiency:** Does LGSR require fewer demonstrations to achieve competitive performance?
- **Robustness:** How does LGSR perform under constrained information settings (e.g., reduced input or smaller representation capacity)?
- **Component Analysis:** How do different components of the task instruction guidance contribute to performance?
- **Representation Quality:** Are the learned LGSR representations better aligned with task-relevant semantics?

A. Experimental Setup

1) Benchmark Environment: We evaluate our method on the CALVIN benchmark [15], a challenging simulation suite designed for language-conditioned policy learning in long-horizon robot manipulation. The benchmark comprises four distinct environments (A, B, C, D), each with a unique object layout and appearance, and includes over 400 natural language instructions spanning 34 manipulation tasks. These tasks are inherently ambiguous from vision alone, making CALVIN particularly well-suited for evaluating multi-task policy learning conditioned on language instructions. This choice is further supported by the task-ID probe in Table I, where Seer reaches 48.54% on CALVIN but 99.62% on LIBERO-Long [25], indicating greater task ambiguity in CALVIN.

In our experiments, we adopt the most challenging CALVIN setup, training on environments A, B, and C and exclusively testing on the unseen environment D. We follow the long-horizon multi-task language control (LH-MTLC) evaluation protocol introduced in [15], where each evaluation sample consists of a sequence of 5 subtasks sampled from the 34 available tasks. The goal of each subtask is defined by a language instruction, and the final state of one subtask becomes the initial state of the next. A total of 1000 long-horizon episodes are evaluated, with the agent executing subtasks sequentially until failure.

2) Baselines for Comparison: To evaluate the effectiveness of our approach, we compare it against several previous methods that have demonstrated competitive performance on the CALVIN benchmark. MCIL [24] is the official baseline that learns a shared latent space between language and image goals. Roboflamingo [26] leverages vision-language foundation models to improve policy learning. SuSIE [29], CLOVER [30], and GHIL-Glue [31] adopt a two-stage paradigm, predicting future observations and inferring actions through the inverse dynamics model. GR-1 [32] uses generative video pre-training, while GR-MG [33] incorporates multimodal goals to utilize diverse training signals. OpenVLA [44] is a representative vision-language-action model that fine-tunes a 7B vision-language backbone on Open X-Embodiment dataset [45]. Seer [14] introduces a tightly coupled language-conditioned world model that predicts future states from task instructions and generates actions accordingly. While Seer demonstrates the promise of integrating task intent into world models, it additionally leverages language-free trajectories for pre-training and mainly uses language for future-state prediction and action generation, rather than explicitly shaping the state representation. In contrast, our method focuses on learning language-guided state representations from language-annotated demonstrations, steering the world model toward task-relevant features through encoder-stage language guidance.

3) Evaluation Metrics: In our framework, the world model regularizes representation learning for the downstream policy and is not deployed as a generative simulator at inference. Therefore, we evaluate the model based on task execution performance, which directly reflects the effectiveness of the learned representations for policy learning. Specifically, we utilize the success rate as the primary metric to evaluate the performance of the learned multi-task policy on individual tasks. For long-horizon tasks, we report the average number of consecutively completed subtasks within each long-horizon instruction sequence (Avg. Len.). All results are averaged over the top three checkpoints based on validation performance to ensure stability and fairness in comparison.

4) Implementation Details: All visual observations are resized to 224×224 and encoded by a frozen pre-trained MAE vision encoder (ViT-B/16) [6]. Task instructions are tokenized and encoded by a frozen pre-trained CLIP text encoder (ViT-B/32) [41]. Proprioceptive inputs, including end-effector position, orientation, and gripper status, are processed by a two-layer MLP with a hidden dimension of 384. The MPE module uses a perceiver resampler [42] with three interleaved cross-attention and feed-forward layers, 6 learnable latent tokens, and a hidden dimension of 768. The reward head uses one interleaved self-attention and cross-attention layer, one learnable latent token, and a hidden dimension of 384. The world-model backbone contains 24 transformer layers with a hidden dimension of 384, the vision decoder has 2 transformer layers, and the action head is a three-layer MLP with a hidden dimension of 192. We set history length $H = 10$, future observation step $N = 3$, action chunk length $M = 3$, and loss weights $w_{\text{disc}} = 0.01$, $w_a = 1.0$, $w_f = 0.1$, and $w_r = 0.01$. w_{disc} and w_f are chosen by matching the gradient scale of future-observation reconstruction to the action loss, while w_r is selected by a small grid search. The model has 320M parameters, of which 68.6M are trainable. Training takes approximately 192 GPU hours on NVIDIA RTX 4090 GPUs, and inference runs at approximately 40 Hz.

B. Experimental Results

1) Comparison with Baselines: As shown in Table II, our method achieves an average sequence length (Avg. Len.) of 4.22, significantly outperforming previous approaches on the CALVIN ABC-D benchmark. Notably, it achieves a 71.9% success rate on the most challenging metric of completing five-step long-horizon tasks, demonstrating strong capability in complex sequential decision-making.

To further evaluate the impact of language-guided representation learning, we compare LGSR with Seer, a recent world model-based method that does not incorporate language during state representation learning. LGSR outperforms Seer (scratch) in terms of average sequence length and achieves competitive performance compared to Seer, which benefits from additional pre-training on unlabeled data. With the introduction of proxy reward prediction, LGSR improves to an average sequence length of 4.22, yielding a 15.9% gain over Seer (scratch) and a 6% gain over Seer (pre-trained), using only language-annotated data from CALVIN. These results underscore the importance of integrating language guidance in world model state representation learning.

We also compare LGSR against GR-MG, a method that leverages large-scale multimodal goal pre-training and strong model backbones to enhance generalization. While GR-MG achieves slightly better results on single-step tasks, LGSR consistently outperforms it on multi-step task completions (lengths 2–5) and in overall sequence length. Importantly, LGSR uses only 68.6M trainable parameters (roughly 8% of GR-MG and SuSIE, and 7% of Roboflamingo) and is trained solely on the CALVIN language split, in contrast to GR-MG's reliance on large-scale pre-training with the Ego4D dataset [46]. This comparison highlights the efficiency and practicality of LGSR in low-resource settings.

We further compare LGSR with OpenVLA, which adapts a frozen 7B vision-language backbone via LoRA-rank-32 adapters and is pretrained on Open X-Embodiment. As shown in Table II, OpenVLA obtains an Avg. Len. of 3.27 on CALVIN ABC-D, while LGSR reaches 4.22 with only 68.6M trainable parameters and no robot-data pretraining. This suggests that LGSR can achieve superior performance over VLM-based policy even without robot-scale pretraining data.

Beyond large-scale pre-trained models, we also evaluate LGSR against two-stage frameworks such as CLOVER and GHIL-Glue. At a comparable trainable size, LGSR exceeds GHIL-Glue by 0.53 in Avg. Len. without depending on a robot-video-pretrained image-editing backbone, and surpasses CLOVER under the same data budget, further demonstrating the data efficiency and effectiveness of our method.

2) Effect of the Proxy Reward Form: To evaluate the proxy reward design, we compare a binary success signal, elapsed progress, and decreasing time-to-go under identical settings. As shown in Table III, the time-to-go target achieves an Avg. Len. of 4.22, outperforming elapsed progress at 3.93 and binary success at 3.76. We attribute these differences to how each target distributes supervision along the trajectory. The binary target provides sparse terminal feedback, while the elapsed-progress target emphasizes later states with weaker task-specific cues. In contrast, the time-to-go target focuses more on earlier states, where object configurations and target identities offer stronger task-discriminative information.

3) Data Efficiency: To evaluate whether LGSR achieves competitive performance with only limited training data, we investigate its performance under varying proportions of training data, specifically 10%, 20%, 40%, 70%, and 100%, which are randomly sampled from the original training dataset. We compare LGSR against its variant without language guidance (denoted as LGSR (w/o Lang.)) and Seer [14] (finetuned on the same fractions of $\mathcal{D}_l$ on top of its full-$\mathcal{D}_{nl}$ pretraining) using the average sequence length (Avg. Len.) as the evaluation metric. As illustrated in Fig. 2(a), LGSR significantly outperforms LGSR (w/o Lang.) once the available data exceeds 20%, demonstrating the importance of language-guided state representations in improving performance. Notably, LGSR achieves comparable performance to its full-data counterpart using only 70% of the training data, indicating strong data efficiency. Furthermore, LGSR trained on 40% of the data surpasses LGSR (w/o Lang.) trained with 70%, further underscoring the benefits of incorporating language as guidance for efficient representation learning. The external comparison with Seer further supports this conclusion: LGSR with only 70% of $\mathcal{D}_l$ and no $\mathcal{D}_{nl}$ already exceeds Seer trained with full $\mathcal{D}_l + \mathcal{D}_{nl}$, while Seer retains an advantage in the very-low-data regime ($\leq 20\%$) due to its $\mathcal{D}_{nl}$ pretraining fallback. These findings suggest that LGSR requires fewer demonstrations to reach competitive performance, which is particularly valuable in low-data or high-cost data collection scenarios.

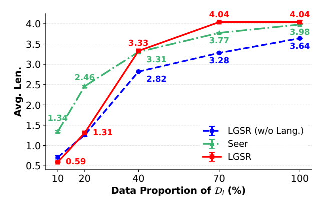
4) Robustness under Limited Representation Capacity: To evaluate the robustness of LGSR under constrained information settings, we investigate its performance when varying the number of state representation tokens from 1 to 6. We report the average sequence length (Avg. Len.) as the evaluation metric. As shown in Fig. 2(b), LGSR consistently outperforms its non-language variant (LGSR w/o Lang.) across all token numbers, demonstrating that the benefits of language-guided representation learning persist even under severe capacity bottlenecks. Notably, with only a single token, LGSR achieves a 17.7% relative improvement over LGSR (w/o Lang.), indicating that language guidance enables more compact and informative state representations. Additionally, LGSR exhibits smaller fluctuation across different token settings, suggesting greater stability and robustness in representation learning when

TABLE II
CALVIN ABC-D RESULTS. $N_{\text{trainable}}$ REFERS TO THE TRAINABLE PARAMETERS. $\mathcal{D}_l$ AND $\mathcal{D}_{nl}$ REPRESENT THE SUBSETS OF THE CALVIN ABC DATASET WITH AND WITHOUT LANGUAGE ANNOTATIONS. $\mathcal{D}_e$ AND $\mathcal{D}_{\text{oxe}}$ DENOTE THE EGO4D AND OPEN X-EMBODIMENT DATASETS.

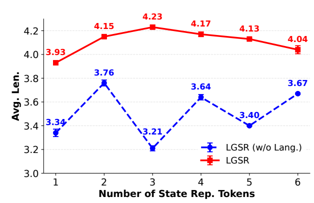
Method	No. Instructions in a Row (1000 chains)						$N_{\text{trainable}}$	Training Data			
	1	2	3	4	5	Avg. Len.		$\mathcal{D}_l$	$\mathcal{D}_{nl}$	$\mathcal{D}_e$	$\mathcal{D}_{\text{oxe}}$
MCIL	21.1%	1.3%	0.0%	0.0%	0.0%	0.22	73M	✓	✓		
Roboflamingo	82.4%	61.9%	46.6%	33.1%	23.5%	2.48	∼1B	✓			
SuSIE	87.0%	69.0%	49.0%	38.0%	26.0%	2.69	∼885M	✓	✓		
GR-1	85.4%	71.2%	59.6%	49.7%	40.1%	3.06	46M	✓		✓	
CLOVER	96.0%	83.5%	70.8%	57.5%	45.4%	3.53	∼200M	✓			
GHIL-Glue	95.2%	88.5%	73.2%	62.5%	49.8%	3.69	∼70M	✓	✓		
GR-MG	**96.8%**	89.3%	81.5%	72.7%	64.4%	4.04	∼900M	✓		✓	
OpenVLA	91.3%	77.8%	62.0%	52.1%	43.5%	3.27	∼100M	✓			✓
Seer (scratch)	93.0%	82.4%	73.0%	62.6%	53.3%	3.64	65M	✓			
Seer	94.4%	87.2%	79.9%	72.2%	64.3%	3.98	65M	✓	✓		
LGSR	95.1%	88.4%	81.1%	73.2%	66.0%	4.04	65M	✓			
LGSR (w/ proxy reward)	96.3%	**90.8%**	**84.5%**	**78.1%**	**71.9%**	**4.22**	68.6M	✓			

TABLE III
PROXY-REWARD ABLATION ON CALVIN ABC-D.

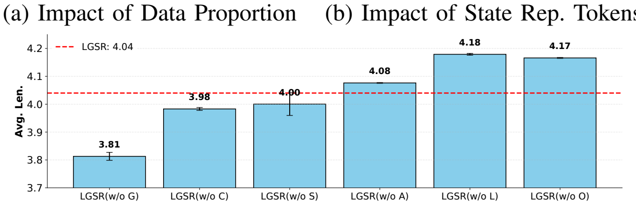
Target	1	2	3	4	5	Avg. Len.
Binary success $r_t = \mathbf{1}[t = T]$	92.4%	83.5%	74.6%	66.5%	58.6%	3.76
Elapsed progress $r = t/T$	94.8%	87.0%	78.8%	70.2%	62.8%	3.93
Time-to-go $r = (T - t)/T$	**96.3%**	**90.8%**	**84.5%**	**78.1%**	**71.9%**	**4.22**



(a) Impact of Data Proportion



(b) Impact of State Rep. Tokens



(c) Impact of Task Instruction Guidance Components

Fig. 2. Ablation Study. (a) Impact of data proportion on performance, where 100% corresponds to 17,870 language-annotated episodes. (b) Impact of the number of state representation (Rep.) tokens on performance. (c) Impact of different components in task instruction guidance on performance. G: Goal; C: Constraint; S: Spatial; A: Action; L: Logic; O: Other. The error bars denote standard error of the mean (SEM).

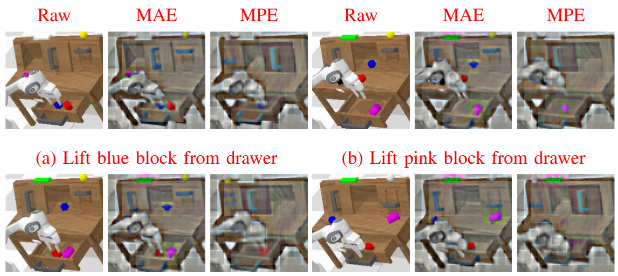
conditioned on task instructions. These results confirm that incorporating language not only enhances the expressiveness of the learned state but also improves performance stability under restricted representation capacity.

5) Component Analysis of Task Instruction Guidance: To assess the contribution of instruction components in state representation learning, we categorize words into five policy-relevant groups: Goal (G), Constraint (C), Spatial (S), Action (A), and Logic (L), with others labeled as Other (O). For example, ‘grasp the red block and rotate it right’ is labeled as ‘A O C G L A G S’. The annotations are first generated by GPT-5 [47] and then verified by human annotators. We conduct category-wise masking on instructions used for representation guidance, while keeping policy-learning instructions unchanged. This setup isolates the effect of instruction components on representation learning, since the policy is always conditioned on the full instruction during inference. As shown in Fig. 2(c), masking goal, constraint, and spatial words substantially degrades performance, showing that object-defining words are crucial for task-relevant representation learning. Without these cues, the multimodal perceiver encoder (MPE) struggles to ground task-relevant objects and associate visual cues with task goals. In contrast, masking action words has little effect, while removing logic and other words can improve performance, suggesting that filtering redundant or distracting words helps focus representation learning on task-relevant features.

6) Qualitative Analysis of Task-Relevant Representations: To better understand how Language-Guided State Representation (LGSR) facilitates policy learning, we compare reconstructions from frozen MAE vision-encoder features and the one-token LGSR produced by MPE. The MAE features are reconstructed using the frozen MAE decoder. For the MPE pathway, we train an inverse decoder, which is the only learned component in this analysis, to project the frozen one-token LGSR back to MAE features before applying the same frozen MAE decoder. As shown in Figure 3, the MPE reconstructions retain task-relevant scene elements while suppressing irrelevant visual content. Specifically, panels (a)–(c) preserve the blue, pink, and red target blocks, respectively, while panel (d) preserves the drawer as the central interaction object. Compared with the MAE reconstructions, the MPE outputs discard more fine-grained appearance details but retain task-relevant objects. These results indicate that MPE compresses multimodal inputs into an LGSR that prioritizes task-relevant scene content, thereby facilitating more effective policy learning.

VI. CONCLUSION

In this work, we propose Language-Guided State Representation (LGSR) learning for world models, which uses natural language task descriptions as semantic priors to encode task-relevant information into latent states. By aligning internal representations with high-level task intent, LGSR enables



(a) Lift blue block from drawer (b) Lift pink block from drawer

(c) Lift red block from drawer (d) Close drawer

Fig. 3. Qualitative visualization of task-relevant reconstructions.

more effective multi-task policy learning. Experiments on CALVIN show consistent gains in long-horizon performance, strong data efficiency with limited data, and robustness under constrained information. Component analysis highlights the role of object-defining words in task-relevant feature extraction, while bottleneck reconstructions show that LGSR focuses on semantically meaningful regions. However, our method also has several limitations. The reward head is trained only on successful demonstrations, leaving its behavior on failure rollouts underexplored. The time-to-go target requires either clip-level alignment between language and trajectories or task-specific milestones, and LGSR still depends on costly language-annotated expert demonstrations, and its benefits depend on task characteristics and benchmark settings. Future work will explore failure-aware supervision, non-expert or language-free demonstrations, annotated human videos, and pre-training strategies for world-model representation learning without dense language annotations.

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